Logarithmic Algebraic Geometry: Exercise Sheet I

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Exercise 1 (12 points)**.** Recall that a monoid P is called *toric* if it is finitely generated, integral, saturated, and if we have $P^{\mathrm{gp}} \cong \mathbb{Z}^{\oplus r}$ for some $r \geq 0$. In this exercise, we show that P is toric if and only if $P \cong P_\sigma$ for some strongly convex rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$.

(a) Let $\widetilde{M} = P^{\mathrm{gp}}$ and $\widetilde{M}_{\mathbb{R}} = \widetilde{M} \otimes_{\mathbb{Z}} \mathbb{R}$. Show that $\tau \coloneqq \mathbb{R}_{\geq 0} \cdot P \subseteq \widetilde{M}_{\mathbb{R}}$ is a full-dimensional rational polyhedral cone.

(b) Show that $\sigma \coloneqq \tau^\vee$ is a strongly convex rational polyhedral cone in $N_{\mathbb{R}} \coloneqq \mathrm{Hom}_{\mathbb{R}}(\widetilde{M}_{\mathbb{R}}, \mathbb{R})$.

(c) Let $M = \mathrm{Hom}_{\mathbb{Z}}(N, \mathbb{Z})$ and $P_\sigma = \sigma^\vee \cap M$. Show that $P \cong P_\sigma$.

Exercise 2 (12 points)**.** Let X be a scheme, and let $\mathcal{M} \to \mathcal{O}_X$ and $\mathcal{N} \to \mathcal{O}_X$ be two log structures. Let $h^\sharp \colon \mathcal{M} \to \mathcal{N}$ be a homomorphism of sheaves of monoids which induces a morphism of log schemes $h \colon (X, \mathcal{N}) \to (X, \mathcal{M})$. We say that $h^\flat$ is a *homomorphism of log structures*. Assume that $h^+ \colon \overline{\mathcal{M}} \to \overline{\mathcal{N}}$ is an isomorphism, and that $\mathcal{M}$ and $\mathcal{N}$ are sheaves of integral monoids. Show that $h^\flat \colon \mathcal{M} \to \mathcal{N}$ is an isomorphism. It is not enough to refer to the related fact about strict morphisms stated in the lecture.

Exercise 3 (14 points)**.** Let $X = \mathbb{P}^1$, $D = \{0, \infty\}$, and $(X, \mathcal{M}_X) = (X|D)$.

(a) Compute $\Gamma(X, \overline{\mathcal{M}}_X)$.

(b) Let $\varpi \colon \mathcal{M}_X \to \overline{\mathcal{M}}_X$ be the quotient map. For each $p \in \Gamma(X, \overline{\mathcal{M}}_X)$, compute

$$\mathcal{L}_X^\times(p) \coloneqq \varpi^{-1}(p)$$

as a subsheaf of $\mathcal{M}_X$. *Hint:* Start by showing that $\mathcal{L}_X^\times(p)$ is locally isomorphic to $\mathcal{O}_X^\times$.

Exercise 4 (18 points)**.** Let $X^\dagger = (X, \mathcal{M}_X)$ be a log scheme, and let $\beta \colon P \to R$ be a pre-log ring. Let $\Gamma(X^\dagger) = (\Gamma(X, \mathcal{M}_X) \to \Gamma(X, \mathcal{O}_X)) \in \mathrm{CRing}^{\mathrm{plog}}$. Construct the bijection

$$\mathrm{Mor}_{\mathrm{Sch}^{\log}}(X^\dagger, \mathrm{Spec}(P \to R)) \cong \mathrm{Hom}_{\mathrm{CRing}^{\mathrm{plog}}}(P \to R, \Gamma(X^\dagger))$$

explicitly. Show that your construction does indeed yield a bijection.

Exercise 5 (24 points)**.** In this exercise, we investigate the failure of $\mathrm{Spec} \colon (\mathrm{CRing}^{\log})^{\mathrm{op}} \to \mathrm{Sch}^{\log}$ to be fully faithful more closely. We consider the pre-log ring

$$(\beta \colon P \to R) = (\beta \colon \mathbb{N} \oplus \mathbb{C}^\times \to \mathbb{C}[t])$$

with $\beta(n, \lambda) = \lambda \cdot t^n \cdot (t-1)^n$. We write $X = \mathrm{Spec}\, R = \mathbb{A}^1_t$.

(a) Show that $(P \to R)$ is a log ring, not only a pre-log ring.

(b) Let $\mathcal{P}_X = (\underline{P}_X)^{\log}_{\beta}$ be the logification of the constant pre-log structure $\underline{P}_X \to \mathcal{O}_X$, and let $\mathcal{M}_X = \mathcal{M}_{(X|\{0,1\})}$ be the divisorial log structure defined by $\{0,1\} \subset X$. Why do we have a comparison map $\delta\colon \mathcal{P}_X \to \mathcal{M}_X$?

(c) Compute the ghost sheaf $\overline{\mathcal{P}}_X$ by studying the construction of the logification and the ghost sheaf.

(d) Compute the ghost sheaf $\overline{\mathcal{M}}_X$. Here, it is not sufficient to refer to the fact that $A_P^{\dagger} \cong (A_P|D_P)$ for $P = \mathbb{N}$. Instead, study functions on X directly.

(e) Show that the induced map $\delta_+\colon \overline{\mathcal{P}}_X \to \overline{\mathcal{M}}_X$ is an isomorphism. Conclude that $\delta\colon \mathcal{P}_X \to \mathcal{M}_X$ is an isomorphism.

(f) Compute $\widetilde{P} \coloneqq \Gamma(X, \mathcal{P}_X) = \Gamma(X, \mathcal{M}_X)$ explicitly.

(g) Show that $(\underline{\widetilde{P}}_X)^{\log} \to \mathcal{P}_X$ is an isomorphism. Conclude that

$$\operatorname{Spec}(\widetilde{P} \to R) \to \operatorname{Spec}(P \to R)$$

is an isomorphism. Explain why this implies that $\operatorname{Spec}\colon (\mathrm{CRing}^{\log})^{\mathrm{op}} \to \mathrm{Sch}^{\log}$ is not fully faithful.

Exercise 6 (20 points)**.** In this exercise, we show that the formation of the divisorial log structure does not always commute with pullback of log structures along étale morphisms. This is one of the reasons why log structures are often considered in the étale topology rather than the Zariski topology (as we do in this lecture).

Let $X = \operatorname{Spec}\mathbb{C}[x,y]$, and let $D = \{y^2 - (x^2 - x^3) = 0\}$. This is a nodal elliptic curve. Let $(X, \mathcal{M}_X) = (X|D)$, and let $o \in X$ be the point given by $\{o\} = \{x = y = 0\}$.

(a) Show that $\overline{\mathcal{M}}_{X,o} \cong \mathbb{N}$. To do so, determine which functions $r \in \Gamma(W, \mathcal{O}_X)$ in a neighborhood $o \in W \subseteq X$ have the property that $r|_{W \setminus D}$ is invertible.

Now, let $Y = \operatorname{Spec}\mathbb{C}[y, t^{\pm}]$, and let $\phi\colon Y \to X$ be given by $\phi^{\sharp}(x) = 1 - t^2$ and $\phi^{\sharp}(y) = y$. Let $E = \phi^{-1}(D) = \{y^2 - (1-t^2)^2 t^2 = 0\} \subset Y$, and let $(Y, \mathcal{M}_Y) = (Y|E)$.

(b) Show that $\phi\colon Y \to X$ is étale.

(c) Show that we have an induced morphism of log schemes $\phi\colon (Y|E) \to (X|D)$.

Let $p \in Y$ be the point given by $\{p\} = \{y = 0, t = 1\}$. We have $\phi(p) = o$.

(d) Show that $\overline{\mathcal{M}}_{Y,p} \cong \mathbb{N}^{\oplus 2}$. Conclude that $\phi\colon (Y|E) \to (X|D)$ is not strict.